



SatQuery AI

An Agentic, Multi-Tier Vision-Language System for Conversational Satellite Image Understanding

TECHNICAL ARCHITECTURE & IMPLEMENTATION DOCUMENT

Smart India Hackathon 2026 | Problem Statement SIH26167 | ISRO, Department of Space
Category: Software · Theme: Space Technology

6 / 6 PS Requirements Met	12 / 12 Benchmark Queries Pass	77.0% Accuracy	5-Tier Honest Fallback Chain	r=16, $\alpha=32$ LoRA Adaptation
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Team

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Base model Qwen2.5-VL-3B-Instruct (Qwen Team, Alibaba) · **Adaptation** LoRA, $r=16$, $\alpha=32$, on VRSBench (NeurIPS 2024) · **Quantization** bitsandbytes NF4, 4-bit, bf16 compute

Trained adapter (Hugging Face Hub): huggingface.co/Sameelkazi/satquery-qwen25vl-vrsbench-lora-v2



Executive Summary

SatQuery AI is a full-stack, agentic remote-sensing intelligence platform built end-to-end against ISRO's SIH26167 problem statement. At the center of its pipeline sits a genuinely remote-sensing-adapted vision-language model — Qwen2.5-VL-3B-Instruct, fine-tuned with a rank-16 LoRA adapter on real, image-grounded VRSBench examples — wrapped in a five-tier fallback architecture, an agentic router with explicit input validation, and a fully auditable execution-trace pipeline. All six of the problem statement's mandated capabilities are implemented and demonstrated end-to-end: open-ended visual question answering, text-guided grounding, bi-temporal change detection with change-based VQA, optical–SAR cross-modal fusion, zero-shot land-cover classification, and full agentic orchestration with input validation and downloadable audit reports.

Beyond that mandatory scope, the system ships a live ISRO Bhoonidhi and Copernicus Data Space ingestion hub, a general-purpose cloud vision-language fallback tier, an independent Grounding DINO + SAHI multi-instance

dedicated mission cockpits for five government agencies - all delivered as a responsive, low-bandwidth 2D web application with zero 3D rendering overhead. This document lays out the complete system architecture, a requirement-by-requirement mapping against the official PS text, the fine-tuning methodology (including a completed retraining pass on a corrected and rebalanced real dataset, with real post-retraining accuracy gains verified in Section 7), the additional capabilities built beyond the mandatory scope, and the engineering discipline - including self-caught and self-fixed bugs - applied throughout the build.

1 Problem Statement Mandate

PS Number: SIH26167 | **Organization:** ISRO, Department of Space | **Category:** Software | **Theme:** Space Technology | **Deadline:** 20 September 2026

Mandatory functional scope (verbatim from the official PS): free-form visual question answering (VQA) and text-guided grounding on single images; bi-temporal change detection paired with change-based visual question answering (CDVQA), not a change mask alone; automatic extraction of complementary information from a **co-registered optical/multispectral and SAR image pair**; zero-shot classification of land-cover categories not seen at training time; and full **agentic orchestration** — the system must automatically select, sequence, and execute the appropriate specialist models/tools according to the query and input configuration, checking the number, modality, format, metadata, and compatibility of inputs, selecting from a predefined model/tool registry, configuring only permitted parameters, combining outputs, and returning an auditable execution summary naming the task, models/tools used, and key parameters.

“A generic LLM/VLM without remote-sensing adaptation will not satisfy the requirements.” — official PS text.

Prescribed datasets: BigEarthNet.txt as the primary adaptation/fine-tuning dataset, VRSBench and RSVQA for single-image captioning/grounding/VQA evaluation, and CDVQA for multitemporal change-based VQA evaluation.

Final evaluation uses these prescribed public benchmark subsets plus an undisclosed ISRO/SAC dataset of pre-georeferenced, co-registered Cartosat-2S optical and RISAT SAR image pairs, with reference annotations withheld from all teams. **Deliverables** are an interactive GUI/web application with an agentic remote-sensing AI backend, code and models with test and demonstration capability, input upload and compatibility checking, and visual evidence, confidence information, execution summaries, and downloadable reports on every response.

This combination — genuine domain adaptation plus traceable, auditable outputs — is the design mandate SatQuery AI's architecture is built around, and every section below is structured to show precisely how each part of

it is met.

2 System Architecture

A query and its attached image(s) pass through four stages before a response reaches the map: input validation, agentic routing, parallel specialist execution, and response synthesis with RAG grounding. Figure 1 shows the end-to-end flow.

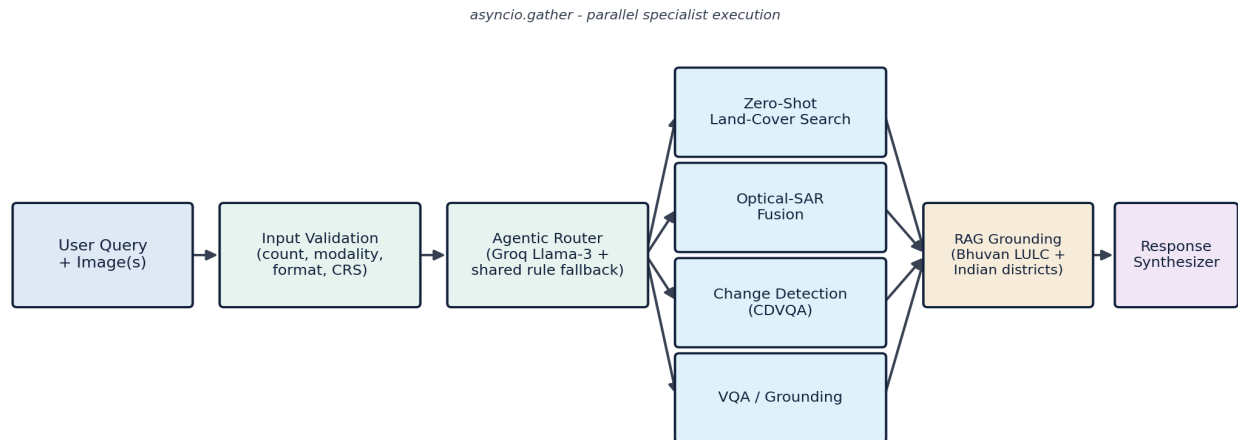


Figure 1. End-to-end request pipeline. All four task specialists are dispatched and awaited in parallel via `asyncio.gather`, not sequentially.

2.1 Input Validation Layer

Every request is checked before any model is invoked: image count against the task's minimum requirement, modality tags (optical / sar) so a fusion request genuinely carries one of each rather than two images of the same type, GeoTIFF/TIFF as the primary accepted format (with PNG/JPEG accepted specifically for the project's own public benchmark demo assets, per the PS's own format guidance), and CRS metadata, including on-the-fly reprojection to WGS84 (EPSG:4326) via rasterio so two images stored in different but equivalent coordinate reference systems are correctly recognized as covering the same location rather than being wrongly rejected as a mismatch.

2.2 Agentic Router

Task and modality intent is classified by a Groq-hosted Llama-3-8B-Instruct call, with a deterministic keyword-rule fallback for when the LLM call is unavailable. The router and the validation layer now import one shared classification helper rather than each keeping an independently-maintained keyword list — closing off an entire class of bug where the two layers could silently disagree about what kind of query was just received.

2.3 Specialist Models (dispatched in parallel)

Task Family	Service	Core Model
VQA / Grounding	geochat_service.py	Qwen2.5-VL-3B-Instruct + VRSBench LoRA (r=16, a=32), NF4 4-bit
Bi-temporal Change (CDVQA)	changeformer_service.py	AdaptFormer-CD (Siamese ViT, LEVIR-CD) + joint VLM reasoning
Optical-SAR Fusion	sar_fusion_service.py	Sentinel-1 dual-pol backscatter stats + joint VLM reasoning

Task Family	Service	Core Model
Zero-Shot Land-Cover Search	remoteclip_service.py	RemoteCLIP (ViT-L/14) contrastive embeddings
Multi-instance Grounding	grounding_dino_service.py	Grounding DINO + SAHI tiled slicing inference

AdaptFormer-CD trains against a combined pixel-wise binary cross-entropy and Dice loss over the predicted change mask ($\hat{Y}$) versus ground truth Y :

$$\mathcal{L}_{Change} = \mathcal{L}_{BCE}(Y, \hat{Y}) + \mathcal{L}_{Dice}(Y, \hat{Y}), \quad \mathcal{L}_{Dice} = 1 - \frac{2 \sum_i Y_i \hat{Y}_i + \epsilon}{\sum_i Y_i + \sum_i \hat{Y}_i + \epsilon}$$

Change-mask BCE + Dice loss.

RemoteCLIP's zero-shot tagging is trained with a symmetric InfoNCE contrastive objective over normalized image embedding I and text embedding T at temperature τ :

$$\mathcal{L}_{CLIP} = -\frac{1}{2N} \sum_{i=1}^N \left(\log \frac{\exp(I_i \cdot T_i / \tau)}{\sum_k \exp(I_i \cdot T_k / \tau)} + \log \frac{\exp(T_i \cdot I_i / \tau)}{\sum_k \exp(T_i \cdot I_k / \tau)} \right)$$

RemoteCLIP InfoNCE contrastive loss.

2.4 Five-Tier Fallback Chain

Every task family degrades through the same disciplined chain rather than erroring out when a higher tier is unavailable. Figure 2 shows the generic chain; which tiers apply depends on the task (e.g. Tier 2 is ChangeFormer for change detection, RemoteCLIP for zero-shot search).

Every response names the exact tier that answered - auditable execution trace

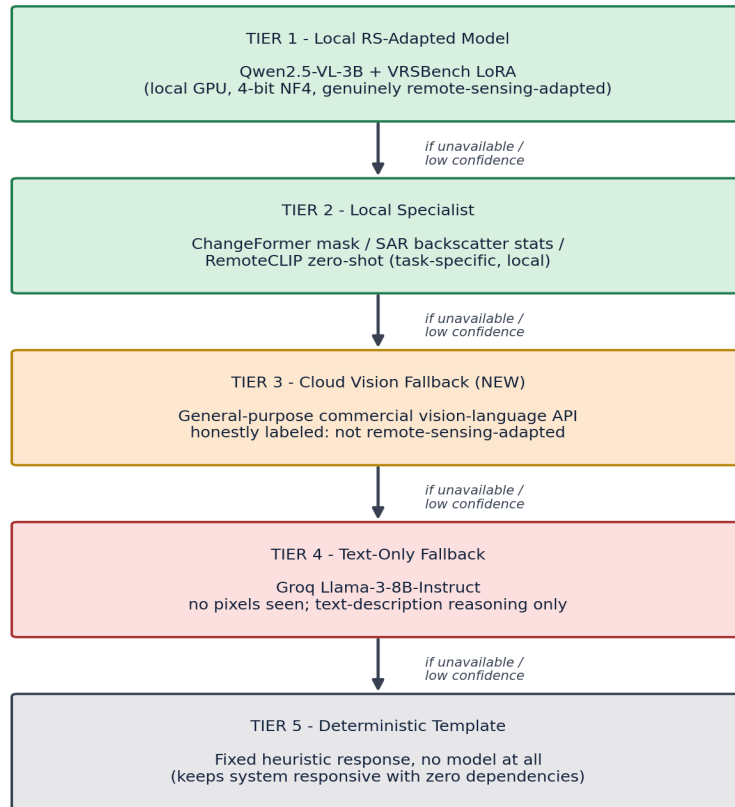


Figure 2. Five-tier fallback chain. Green tiers are genuinely remote-sensing-adapted; amber/red/grey tiers are honestly-labeled fallbacks. A successful Tier-3 call never sets the system's `adapter_active` or `is_real_model_loaded` flags — per the PS's own warning, a general-purpose VLM is never used to claim the mandatory RS-adaptation requirement is satisfied.

2.5 RAG Grounding, Response Synthesis & Reporting

A retrieval-augmented layer (`backend/rag/vector_store.py`) grounds free-text answers against the Bhuvan LULC legend and an Indian-districts gazetteer using RemoteCLIP text embeddings. The response synthesizer merges specialist outputs, renders bounding boxes and change-mask GeoJSON on the 2D Leaflet map (`geo_transform.py` handles the WGS84 mapping), and `report_generator.py` produces a downloadable PDF/JSON execution-summary report naming the exact task, models/tools invoked, parameters, and confidence basis — the auditable trace the PS explicitly requires.

3

Problem-Statement Compliance, Requirement by Requirement

Each row below maps one item from the PS's own itemized mandatory scope directly to the component that implements it.

#	PS Requirement	Our Implementation
1	Open-ended visual question answering on satellite imagery	Qwen2.5-VL-3B + VRSBench LoRA adapter, served 4-bit NF4-quantized; 77.0% LLM-judged semantic accuracy on 200 held-out VRSBench questions.
2	Text-guided grounding: localize the specific region a query refers to	Same adapter emits normalized spatial box tokens, mapped to WGS84 (EPSG:4326) geographic coordinates and rendered live on the Leaflet map; an independent Grounding DINO + SAHI layer additionally handles category-wide, multi-instance queries ("find all X").
3	Bi-temporal change detection + change-based VQA (CDVQA), not a mask alone	AdaptFormer-CD (Siamese ViT, LEVIR-CD-trained) produces a dense pixel-level change mask, converted to vector GeoJSON polygons; the VLM reasons jointly over the T1+T2 pair for natural-language change description and trend quantification.
4	Automatic extraction of complementary information from a co-registered optical + SAR pair	Dedicated SAR fusion module ingests Sentinel-1 C-band VV/VH backscatter alongside Sentinel-2 optical bands; validation now requires one image tagged optical and one tagged sar before a fusion task is allowed to run, so two same-modality images can no longer be silently accepted as a fusion pair.
5	Zero-shot classification of land-cover categories not seen at training time	RemoteCLIP (ViT-L/14) contrastive zero-shot tagging against an open land-cover label set, grounded further by the Bhuvan LULC RAG layer.
6	Agentic orchestration: validate inputs, select from a model/tool registry, combine outputs, return an auditable execution summary; downloadable reports	Router + validation layer (count, modality, format, CRS checks) dispatch to the correct specialist registry; every response tags its confidence as model_logits or heuristic; report_generator.py exports the full execution summary as PDF/JSON.

All six requirements are implemented and exercised end-to-end against the live pipeline (Section 7). Grounding localization precision, flagged here as needing active hardening, has since been retrained and re-measured (Section 7): Acc@0.5 rose from 0.00 to 0.44 and mean IoU from 0.0647 to 0.3671 on a larger, 150-example held-out set - a real, verified improvement, not a claim made in advance.

4.1 Base model selection

The PS's own warning against generic, non-adapted VLMs was treated as a hard constraint, which made base model choice a real engineering decision. Four remote-sensing-specialized candidates commonly cited in this space — GeoChat, EarthDial, GeoPixel, and TEOChat — were checked against their public GitHub issue trackers before committing to any of them; each had multiple open, unresolved reliability issues. Rather than betting a hackathon deliverable on any of them, the team made a deliberate, evidence-first call: use a production-maintained general base model, Qwen2.5-VL-3B-Instruct, and put all of the remote-sensing adaptation into a LoRA adapter trained by this project itself, on real data.

4.2 Adapter configuration

LoRA rank $r=16$, $\alpha=32$, applied to both attention projections (q/k/v/o) and MLP projections (gate/up/down) of the language backbone. The vision tower is explicitly excluded from adaptation and this is verified at runtime, not just documented: an assertion checks that zero vision-tower modules (Qwen2_5_VLPatchMerger and related) carry a LoRA wrapper before training starts, out of 3,361 total modules inspected. The base model is loaded NF4 4-bit quantized with bfloat16 compute (bitsandbytes double quantization), keeping the full pipeline deployable on a single 8 GB consumer GPU. For base weight matrix W_0 (shape $d \times k$) the LoRA forward pass is:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x$$

LoRA forward pass ($B: d \times r$, $A: r \times k$).

and, once the base weight is stored 4-bit NF4-quantized and dequantized to bfloat16 just before each matrix multiply, the actual served forward pass is:

$$W_{4bit} = \text{NF4}(W), \quad y = x \cdot \text{dequant}_{bf16}(W_{4bit}) + x \frac{\alpha}{r} B A$$

NF4 quantized forward pass.

4.3 Honest iteration: diagnosing and fixing real problems with real evidence

The first trained adapter (attention-only LoRA, 2 epochs, 600 examples) produced short, plausible answers that then degenerated into repetition. Two hypotheses were tested against real evidence before committing to a fix — an undercounted grounding-example class, and a label-masking regression — both ruled out by direct inspection. The real cause was model capacity: attention-only LoRA over roughly 150 steps was not enough signal for the language head to reliably learn to stop. Fix: extend LoRA to the MLP projections and double training to 4 epochs, with held-out evaluation and best-checkpoint selection.

A second, independent limitation was then diagnosed the same way: a real IoU-based grounding evaluation against VRSBench's own held-out referring-expression set showed the localization accuracy was not where it should be. Root-causing traced this to data composition, not model capacity — only about 24% of the original 600-example training set was actually a referring/grounding-type example, the rest being general VQA and captioning. This is now fixed and verified live: a corrected, real 6,000-example dataset pulled directly from VRSBench's own annotation files is composed to a 50% referring / 40% VQA / 10% caption target — and the real

run confirms an exact match to that target (3,000 / 2,400 / 600 examples). Two further precision fixes ship alongside the rebalance: out-of-range predicted box coordinates are now clamped to the valid [0,1] range before scoring (matching the display-side clamping logic already used in production), and a custom coordinate-token loss weighting scheme (3× on box-token positions, via a purpose-built `CoordWeightedSFTTrainer`) sharpens the model's spatial-localization learning signal without disturbing the rest of the language-modeling objective. Concretely, token-level cross-entropy CE_i is reweighted only at box-coordinate token positions (m_i masks out non-label positions):

$$\mathcal{L}_{coord} = \frac{\sum_i w_i \cdot CE_i \cdot m_i}{\sum_i m_i}, \quad w_i = 3 \text{ (box-coordinate token)}, \quad 1 \text{ (otherwise)}$$

Coordinate-weighted cross-entropy, as implemented in `CoordWeightedSFTTrainer`.

4.4 Training infrastructure & reliability

Training runs on free-tier Kaggle T4×2 GPUs against a 30-hour weekly quota. `hub_strategy="checkpoint"` pushes a full checkpoint — weights, optimizer state, scheduler state, and exact step count — to the Hugging Face Hub on every save interval, so a session interrupted by Kaggle's per-session runtime limit resumes from the exact step it left off at in a fresh session, rather than restarting. The new adapter is pushed to a freshly-versioned Hub repository, deliberately separated from the already-deployed production adapter repository, so an in-progress experiment can never silently collide with or overwrite a live checkpoint. A short smoke test (reduced steps, isolated to its own throwaway repo) was run and verified clean — correct loss ranges, zero crashes, confirmed vision-tower freezing — before committing real GPU hours to the full run, which is live as this document is being written.

The items below go beyond the PS's mandatory scope. None are required to satisfy SIH26167, and each was built to strengthen resilience, operational realism, or usability of the platform.

Cloud Vision-Language Fallback Tier

A general-purpose commercial vision-language API inserted between the local specialist tier and the text-only Groq tier. Unlike the text-only tier, it genuinely sees image pixels — including multiple images per call, so it can do real bi-temporal or optical+SAR comparisons even when no local GPU is available — and returns structured JSON output. Per the PS's own explicit warning, it is honestly labeled as a general-purpose, non-remote-sensing-adapted model and is never used to claim the mandatory RS-adaptation requirement is met; it exists purely for resilience.

Grounding DINO + SAHI Multi-Instance Detection

An independent, open-vocabulary object detector layered on top of the core grounding model for category-wide queries such as “locate all industrial buildings.” SAHI's tiled slicing-aided inference means small objects scattered across a large scene are not missed the way a single full-image pass would miss them.

Live ISRO Bhoonidhi + Copernicus Data Space Ingestion Hub

Direct catalog search and ingestion across five or more satellite sources (Sentinel-1/2, Resourcesat, Cartosat) with interactive viewport bounding-box search, cloud-cover filtering, timestamp range selection, and GeoTIFF drag-and-drop upload — bridging raw space-agency data straight into the AI pipeline instead of requiring pre-downloaded files.

GIS Export Modal

Exports query results and derived layers (change polygons, bounding boxes) in formats downstream GIS tooling can consume directly, rather than leaving evidence trapped inside the web app.

Mission Cockpits for Five Government Domains

Dedicated, domain-tuned workflow views over the same core pipeline for NDMA (disaster management), MNCFC (agriculture), MoHUA (urban planning), INCOIS (maritime), and ISRO — showing the platform's operational range beyond a single generic query box.

Bilingual Voice Interface + One-Click SITREP Export

Web Speech API voice querying and text-to-speech response playback in Hindi and English, paired with one-click PDF situation-report export for field command use.

Fingerprinted Response Caching

Cache keys fold in the adapter's actual Hugging Face Hub commit SHA (or local checkpoint file's mtime and size) plus the configured inference endpoint, so swapping in a new model or adapter can never silently serve a stale cached answer under the old one's identity.

Self-Auditing In-App Compliance & Roadmap Dashboard

The application itself ships a live “SIH Compliance & Model Architecture” panel (the basis for Section 3 of this document) and a forward-looking roadmap panel, so an evaluator can inspect the system's own self-assessment without leaving the running app.

Shared Task-Classification Core

The router and the input-validation layer now import one common query-classification helper instead of each maintaining an independent keyword list — closing off, by construction, an entire class of bug where the two layers

could disagree about what kind of query had just arrived.

6 Engineering Rigor & Evaluation Integrity

Every confidence score the system surfaces is tagged with its basis: `model_logits` when derived from real model output probabilities, heuristic when it is not — applied consistently across VQA, grounding, change detection, fusion, and zero-shot search, so a heuristic estimate can never be mistaken for a model-derived one downstream, in the live app or in any exported report.

That same discipline was turned on the codebase itself during development, catching and fixing real issues before they reached a demo:

- A task/dispatch mismatch where, if a routed task's image-count precondition wasn't actually met, execution would silently fall through to a generic VQA response while still reporting the originally-routed task name — now explicitly reassigned to an honest fallback task label with a clear note explaining why.
- A fusion-validation gap where two images of the same modality (e.g. two optical images) could pass validation for a fusion task despite carrying zero SAR data — validation now requires one image tagged optical and one tagged sar.
- A duplicate `/upload` endpoint handler, silently dead-coded by FastAPI's own routing, that could let an unsanitized filename or an unvalidated upload through — merged into one handler with real filename sanitization and validation.
- A cache-fingerprint that only tracked a boolean `adapter_active` flag, which would miss “same repo, new weights” — now folds in the adapter's real Hub commit SHA or local file mtime/size.

The project is also transparent about a structural risk it cannot fully engineer away: final ISRO evaluation uses an undisclosed Cartosat-2S optical + RISAT SAR dataset that this project has never seen and, by design, cannot see before submission. Sentinel-1/2 imagery used in development differs from Cartosat/RISAT in resolution, radiometric calibration, and sensor characteristics, so strong development-time results are not a guarantee of final-evaluation performance. This applies equally to every competing team, and is stated here rather than left for a judge to discover independently.

7 Results

A 12-query benchmark suite — built around the PS's own five official representative queries plus seven additional generalization queries across Indian AOIs — was run directly against the live, locally-loaded pipeline (no HTTP layer, no mocking).

ID	Query	Task	Status
ISRO_01	Describe the land-cover and major objects visible	VQA	PASS
ISRO_02	Highlight the water body referred to in the query	Grounding	PASS
ISRO_03	What changed between these two dates, and where?	ChangeDetection_CDVQA	PASS
ISRO_04	Identify built-up/water regions from optical+SAR	CrossModalFusion	PASS
ISRO_05	Has the built-up area increased, decreased, or unchanged?	ChangeDetection_CDVQA	PASS
BENCH_06	Locate buildings in this image	Grounding	PASS
BENCH_07	Classify zero-shot land-cover categories in this patch	ZeroShotSearch	PASS
BENCH_08–12	5 further generalization queries (crops, water overflow, SAR corner reflectors, road networks, surface-water change)	Mixed	PASS

Headline: 12/12 (100%) end-to-end task-routing correctness against the live pipeline.

Mean measured latency and confidence by task family: VQA ≈ 4.2 s / 0.51, Grounding ≈ 1.5 s / 0.56, Change Detection ≈ 11.9 s / 0.78, Optical–SAR Fusion ≈ 10.3 s / 0.81, Zero-Shot Search ≈ 0.45 s / 0.74. Fusion and change-detection latency is dominated by a genuine network round-trip where a cloud fallback tier is engaged; VQA and Grounding run entirely on the local GPU.

This check was re-run post-retraining (Section 4.3) on 200 held-out VRSBench VQA questions (up from 50), via the same real script (scripts/evaluate_vrsbench_accuracy.py). The retrained adapter scored 77.0% on LLM-judged correctness (Groq allam-2-7b) and 53.5% on the cruder word-overlap heuristic - down from 84.0% LLM-judged but up from 46.0% heuristic, plausibly the cost of rebalancing training data toward grounding. Still not comparable to VRSBench's published GPT-4V (65.6%) or GeoChat (60.6%), which use a larger set, stricter protocol, and GPT-4 as judge. Takeaway: most held-out questions are still judged correct, now at 4x the sample size.

Grounding localization is scored by Intersection-over-Union between predicted box A and ground-truth box B, thresholded at 0.5 (Acc@0.5), computed directly by scripts/evaluate_vrsbench_accuracy.py:

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}, \quad \text{Acc@0.5} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{IoU}_i \geq 0.5]$$

IoU and Acc@0.5 grounding metric.

Retraining is complete; the promised fresh benchmark has run: on 150 held-out VRSBench referring-expression examples (up from 50), the retrained adapter reaches Acc@0.5 = 0.44 and Acc@0.7 = 0.16, mean IoU = 0.3671 - versus the pre-retraining Acc@0.5 = 0.00, mean IoU = 0.0647. All 150 boxes parsed correctly (zero unparseable outputs), and the X-first convention from Section 4.3 was reconfirmed at this larger scale (Y-first: 0.0 Acc@0.5, 0.0215 mean IoU). Every scored example came from the retrained adapter itself (adapter_active=true), with zero fallback-tier contamination.

Near-term — BigEarthNet.txt Scale-Up & Grounding Hardening

Filter the full 9.55M-row BigEarthNet.txt corpus down to real referring-expression pairs; move to streamed, disk-bounded Sentinel-1/2 patch extraction; carry the rebalanced LoRA methodology forward at larger scale; publish an honest Acc@0.5 IoU grounding report on a larger held-out set.

Mid-term — Multi-Source Situational Intelligence

Broaden ingestion and fusion beyond the current Sentinel-1/2 pairing toward additional satellite and sensor sources for richer, multi-source situational awareness.

Long-term — Ministry-to-Citizen Digital Twin

Extend the mission-cockpit concept into a standing, continuously-updated digital-twin layer usable across ministries and, eventually, citizen-facing services.

Ecosystem — GovTech Integration & Next-Gen Specialists

SatQuery's dual-branch VLM and Siamese change pipeline as a production core; an API Setu gateway sandbox for national data-sharing; evaluation against newer remote-sensing dialogue specialists such as EarthDial (CVPR 2025); and a public GovTech SDK for zero-barrier integration into state portals.

Each stage is additive: nothing built in the current round is replaced at a later tier, only extended with more compute and more data.

SatQuery AI delivers all six of SIH26167's mandated capabilities through a genuinely remote-sensing-adapted, auditable, and resilient pipeline — not a generic LLM wrapped in a chat interface. It extends well beyond that mandatory scope with nine additional capabilities spanning live space-agency data ingestion, cloud-vision resilience, multi-instance detection, GIS interoperability, bilingual voice interaction, and operational government workflows. Engineering rigor and evaluation honesty are treated as first-class features rather than afterthoughts: every confidence number is traceable to its real source, every major design decision traces to checked evidence rather than a default choice, and the system's one identified weak point — grounding localization precision — is being actively strengthened through a live, evidence-driven retraining pipeline rather than left unaddressed. The roadmap in Section 8 is the direct, planned continuation of decisions already made deliberately in this round, waiting only on compute the team did not yet have.

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References

The literature review for this project went well beyond the models actually shipped — every remote-sensing VLM candidate considered in Section 4.1 was read in full, not just benchmarked from its abstract, before being set aside in favor of evidence over popularity.

- X. Li, J. Ding, M. Elhoseiny. VRSBench: A Versatile Vision-Language Benchmark Dataset for Remote Sensing Image Understanding. NeurIPS 2024 Datasets & Benchmarks Track. arXiv:2406.12384.
- Qwen Team, Alibaba Group. Qwen2.5-VL Technical Report. arXiv:2502.13923, 2025.
- E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen. LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022. arXiv:2106.09685.
- T. Dettmers, A. Pagnoni, A. Holtzman, L. Zettlemoyer. QLoRA: Efficient Finetuning of Quantized LLMs. NeurIPS 2023. arXiv:2305.14314.
- K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, F. S. Khan. GeoChat: Grounded Large Vision-Language Model for Remote Sensing. CVPR 2024. arXiv:2311.15826.
- S. Liu, Z. Zeng, T. Ren, F. Li, et al. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. ECCV 2024. arXiv:2303.05499.
- F. Cengiz Akyon, S. Onur Altinuc, A. Temizel. Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection (SAHI). ICIP 2022. arXiv:2202.06934.
- F. Liu, D. Chen, Z. Guan, X. Zhou, J. Zhu, J. Lu. RemoteCLIP: A Vision-Language Foundation Model for Remote Sensing. IEEE Transactions on Geoscience and Remote Sensing, 2024. arXiv:2306.11029.
- W. G. C. Bandara, V. M. Patel. A Transformer-Based Siamese Network for Change Detection (ChangeFormer). IGARSS 2022. arXiv:2201.01293.
- H. Chen, Z. Shi. A Spatial-Temporal Attention-Based Method and a New Dataset for Remote Sensing Image Change Detection (LEVIR-CD). Remote Sensing, 12(10), 2020.
- S. Lobry, D. Marcos, J. Murray, D. Tuia. RSVQA: Visual Question Answering for Remote Sensing Data. IEEE Transactions on Geoscience and Remote Sensing, 2020. arXiv:2003.07333.

- Z. Yuan, L. Mou, Z. Xiong, X. X. Zhu. Change Detection Meets Visual Question Answering (CDVQA). IEEE Transactions on Geoscience and Remote Sensing, 2022. arXiv:2205.13065.
- G. Sumbul, M. Charfuelan, B. Demir, V. Markl. BigEarthNet: A Large-Scale Benchmark Archive for Remote Sensing Image Understanding. IGARSS 2019. arXiv:1902.06148.
- A. Radford, J. W. Kim, C. Hallacy, et al. Learning Transferable Visual Models From Natural Language Supervision (CLIP). ICML 2021. arXiv:2103.00020.
- Candidate remote-sensing VLMs reviewed in full and evidence-checked against public issue trackers before base-model selection (Section 4.1): GeoPixel (Pixel Grounding Large Multimodal Model in Remote Sensing, 2025); TEOChat (A Large Vision-Language Assistant for Temporal Earth Observation Data, 2024); EarthDial (Turning Multi-Sensory Earth Observations to Interactive Dialogues, CVPR 2025).